TECHNICAL DOCUMENTATION

Service: mailfrom-milter

mailfrom-milter

Postfix milter written in Go that enforces domain alignment between the SMTP envelope sender (`MAIL FROM`) and the `From:` message header for authenticated sessions. Prevents DKIM domain spoofing: an authenticated user setting `MAIL FROM` to their own domain but forging `From:` with a victim domain would produce a valid DKIM signature for a domain they do not own. This milter rejects such messages before DKIM signing occurs.

Only authenticated SMTP sessions (SASL) are checked. Unauthenticated connections (inbound MX delivery) pass through without inspection.

Architecture

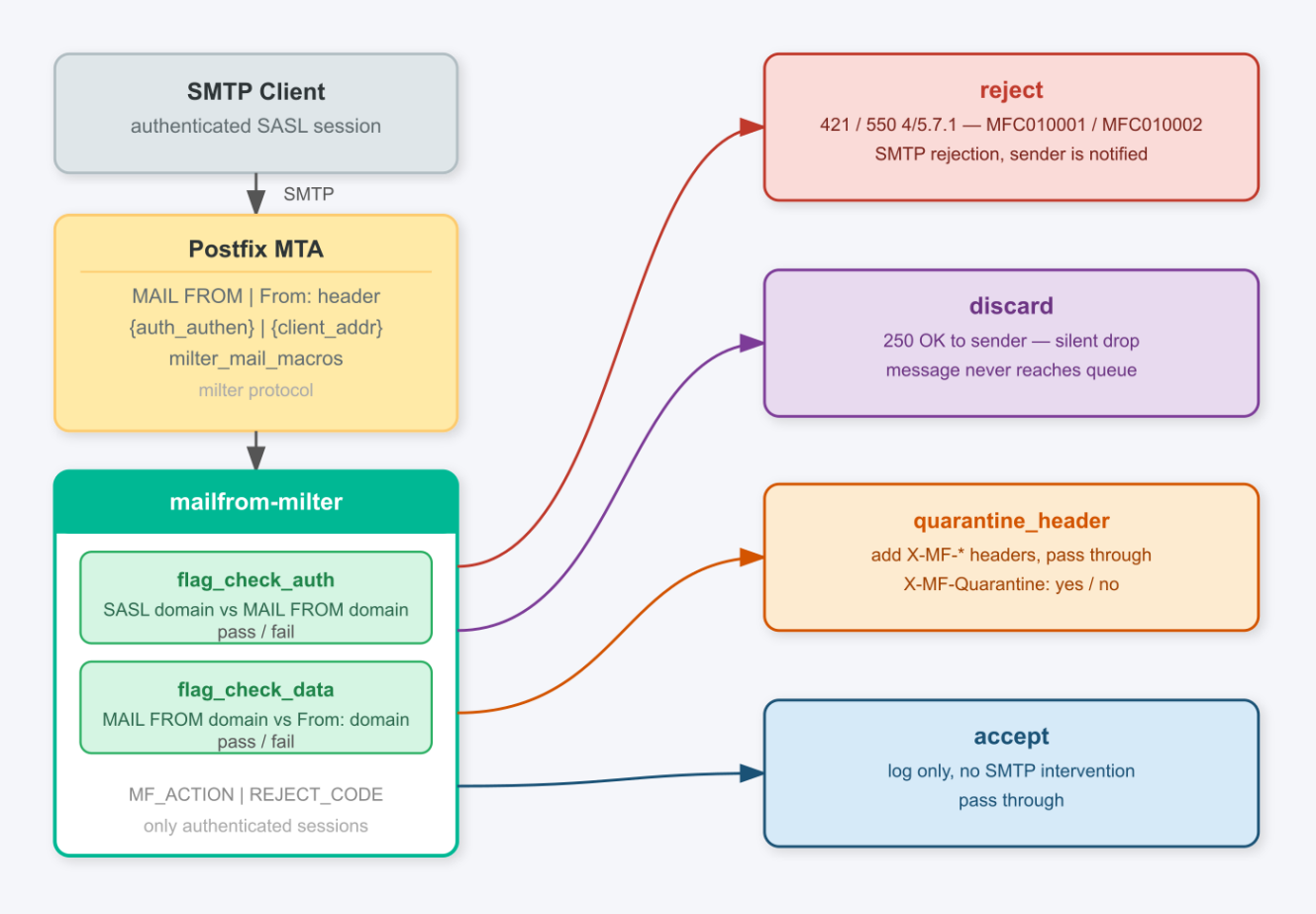


Figure 1: mailfrom-milter architecture

Stack

- Go 1.27 (`alpine:3.24` runtime image)
- `github.com/0kaba0hub/go-milter` v0.5.0 (fork of `emersion/go-milter`, module renamed)
- `log/slog` — structured JSON logging

Directory layout

```
mailfrom-milter/
|- app/go/
|  |- main.go
|  |- Dockerfile
|  |- go.mod
|  \- go.sum
|- helm/
|  |- Chart.yaml
|  |- values.yaml
|  \- templates/
|     |- deployment.yaml
|     \- service.yaml
|- helm_values/
|  \- values-sandbox.yaml
|- doc/
|- argocd-app.yaml
\- .github/workflows/ci.yaml
```

Checks

For every authenticated session two checks are performed:

Flag	Check	Values
`flag_check_auth`	SASL username domain vs `MAIL FROM` domain	`pass` / `fail`
`flag_check_data`	`MAIL FROM` domain vs `From:` header domain	`pass` / `fail`

Actions

Configured via the `MF_ACTION` environment variable:

Action	On check fail	On both pass
`reject`	SMTP reject with `REJECT_CODE` `4/5.7.1`	log + accept
`discard`	silent drop (`250 OK` to sender), log	log + accept
`quarantine_header`	add `X-MF-Quarantine: yes` headers, log	add `X-MF-Quarantine: no` headers, log
`accept`	log only, accept	log only, accept

Headers added for `quarantine_header` action:

Header	Value
`X-MF-Envelope-From`	`MAIL FROM` address
`X-MF-From`	Address extracted from `From:` header

Header	Value
`X-MF-Quarantine`	`yes` if any check failed, `no` if all passed

Environment Variables

Variable	Default	Description
`LISTEN_ADDR`	`0.0.0.0:10031`	TCP address to listen on
`MF_ACTION`	`reject`	`reject` / `discard` / `quarantine_header` / `accept`
`REJECT_CODE`	`421`	SMTP reply code for reject: `421` (temp) or `550` (permanent)
`MILTER_READ_TIMEOUT_SECONDS` / `MILTER_WRITE_TIMEOUT_SECONDS`	`3600` / `60`	milter connection timeouts
`LOG_LEVEL`	—	Set to `debug` for verbose per-message logging

Deploy

Managed by ArgoCD. Apply the ArgoCD Application manifest:

```
kubectl apply -f argocd-app.yaml
```

Postfix configuration:

```
smtpd_milters = inet:mailfrom-milter.mail.svc.cluster.local:10031
milter_mail_macros = i {mail_addr} {client_addr} {client_name} {auth_authen} {auth_type}
milter_default_action = accept
```

Local development:

```
docker build -t mailfrom-milter:dev app/go/
docker run --rm -p 10031:10031 -e MF_ACTION=accept -e LOG_LEVEL=debug mailfrom-milter:dev
```

Logging

Every processed authenticated message produces one JSON log entry:

```
{
  "time": "...", "level": "INFO", "msg": "milter",
  "envelope_from": "user@attacker.com",
  "auth_user": "user@attacker.com",
  "flag_check_auth": "pass",
  "from_header": "ceo@victim.com",
  "flag_check_data": "fail",
  "return_code": "reject"
}
```

`return_code` values: `reject`, `discard`, `accept`. Enable verbose logging without code changes: `LOG_LEVEL=debug`.